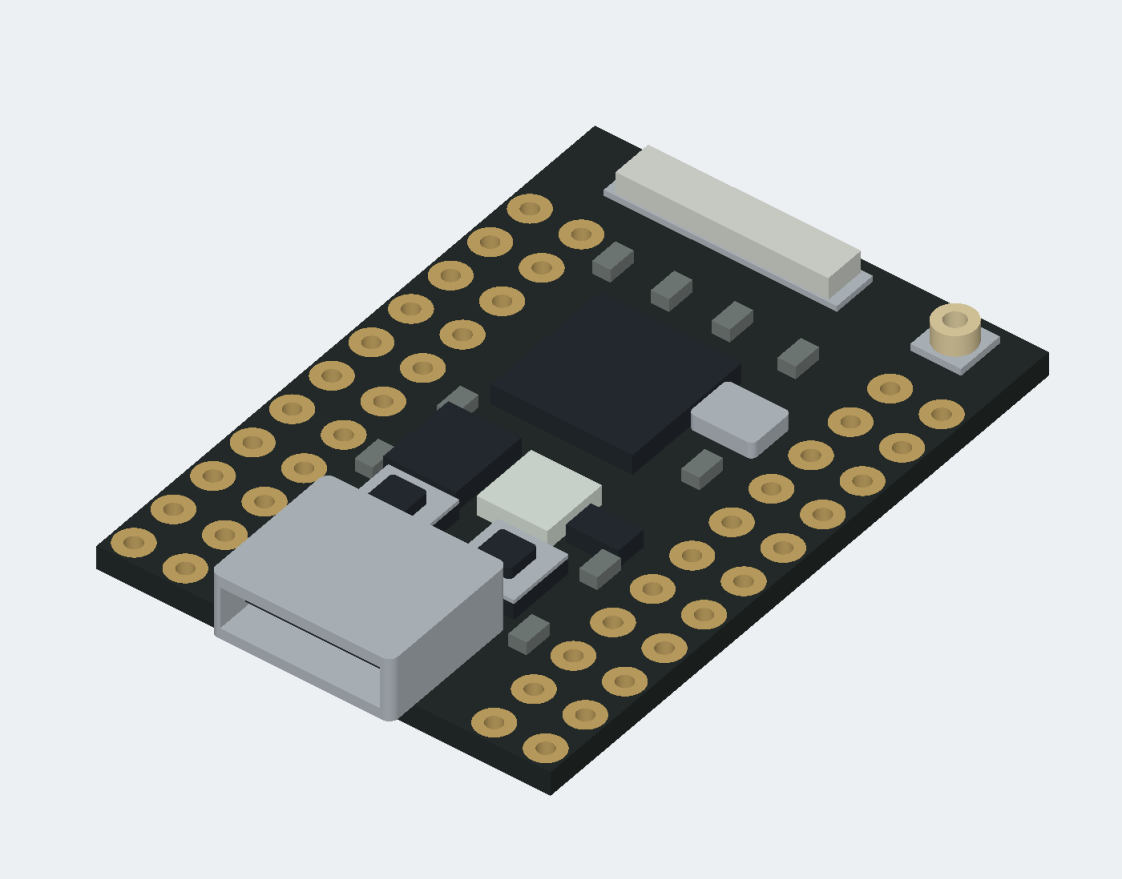
ESP32-S3-N16R8-MiniNano

Compact 44-position development board

Mechanical models and electrical reference for the NiceMCU-branded board in the supplied images. This is not Arduino Nano, Waveshare Nano or the ESP32-S3 SuperMini. Manufacturer schematic and exact revision were not identified; no other board's wiring is substituted.

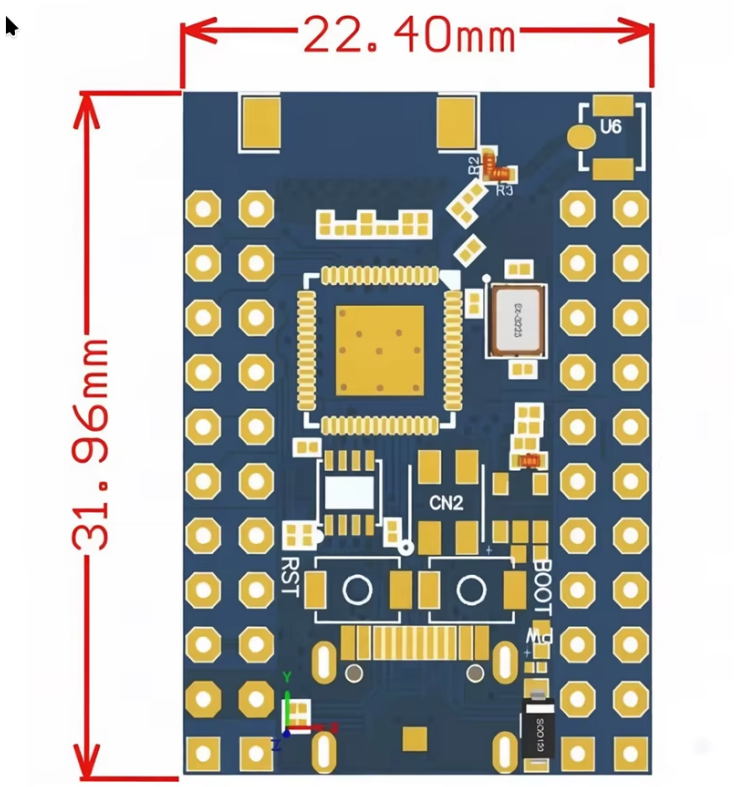


Bare-board model. USB-C is at the near end; ceramic antenna and external RF receptacle at the far end.

Item	Value / scope
Dimensioned PCB	22.40 wide x 31.96 long; estimated thickness 1.20
Header layout	4 columns x 11 positions = 44; inferred 2.54 pitch
Estimated bare envelope	22.40 x 33.46 x 4.40, including assumed USB overhang
Estimated with-header envelope	22.86 x 33.46 x 12.94; header plastic projects slightly beyond PCB sides
Deliverables	Bare and downward-header STEP, LYL, 3MF, STL, previews, pinout and sources

Important: GPIO26 and GPIO33-37 appear in the supplied pinout but overlap flash/PSRAM signals. They must not be assumed available on an N16R8/R8 memory configuration. [S1, S3]

Measured outline, inferred grid and heights



Only 22.40 x 31.96 is explicitly dimensioned. Pin positions and component heights below are inferred.

Feature	Nominal model (mm)	Basis
PCB / thickness	22.40 x 31.96 / 1.20	Outline given; thickness estimated
Longitudinal pitch / span	2.54 / 25.40	11 centers, 10 intervals; drawing grid
Adjacent columns	2.54	Inferred nominal header grid
Inner / outer row gap	15.24 / 20.32	Inferred nominal 2.54 multiples; image supports approximately 15.4 / 20.4
Column centers X	-10.16, -7.62, +7.62, +10.16	A, B, C, D component-side left-to-right
End margins	Antenna 5.52 / USB 1.04	Inferred from drawing center locations
Drill / pad diameter	1.00 / 1.80	Estimated; end square pads simplified as round
USB shell W x L x H	9.00 x 7.50 x 3.20	Estimated above 1.20-thick PCB
USB overhang / total length	1.50 / 33.46	Estimated from photograph; not dimensioned
QFN / flash / crystal height	1.00 / 1.00 / 0.80	Estimated above PCB
Header plastic / pin tail	2.54 / 6.00	Assumed generic 1x11 male strips
Header width / length / pin	2.54 / 27.94 / 0.64 square	Assumed; four strips
Vertical limits, fitted	-8.54 to +4.40	Total estimated height 12.94

Grid derivation: approximate image inner/outer center gaps divided by PCB image width and multiplied by 22.40 give about 15.4/20.4 mm. The selected 15.24/20.32 nominal spacings fit a 2.54 grid within roughly 0.15 mm image uncertainty. Verify before manufacturing a mating PCB.

Layerling coordinates: PCB center X/Z = 0; underside elevation = 0; antenna -Z, USB +Z. Pin centers Z = -10.46 + 2.54 i, i=0...10. STEP is Z-up; STEP Y is opposite Layerling Z.

All 44 pin definitions

Component-side view, antenna at top and USB at bottom. Columns A/B are outer/inner left; C/D are inner/outer right. Positions 01-11 run antenna-to-USB. IDs are assigned for this model, not package pins or Arduino D-number aliases. Mapping is transcribed from the supplied interface drawing.

Row	A outer-left	B inner-left	C inner-right	D outer-right
01	**GPIO0** boot-time strap	**EN** CHIP_PU, active-low reset	**GPIO1** ADC1 CH0; Touch1	**GPIO3** ADC1 CH2; Touch3; boot-time strap
02	**GPIO7** ADC1 CH6; Touch7	**GPIO8** ADC1 CH7; Touch8	**GPIO4** ADC1 CH3; Touch4	**GPIO2** ADC1 CH1; Touch2
03	**GPIO9** ADC1 CH8; Touch9	**GPIO10** ADC1 CH9; Touch10	**GPIO6** ADC1 CH5; Touch6	**GPIO5** ADC1 CH4; Touch5
04	**GPIO11** ADC2 CH0; Touch11	**GPIO12** ADC2 CH1; Touch12	**TX / GPIO43** UART0 TX, GPIO43 [S1]	**GPIO45** boot-time strap
05	**GPIO13** ADC2 CH2; Touch13	**GPIO14** ADC2 CH3; Touch14	**RX / GPIO44** UART0 RX, GPIO44 [S1]	**GPIO42** JTAG MTMS
06	**GPIO15** ADC2 CH4	**GPIO16** ADC2 CH5	**GPIO46** boot-time strap	**GPIO41** JTAG MTDI
07	**GPIO17** ADC2 CH6	**GPIO18** ADC2 CH7	**GPIO39** JTAG MTCK	**GPIO40** JTAG MTDO
08	**GPIO21** General digital I/O; board loads unverified	**GPIO26** **memory caution**	**GPIO36** **memory caution**	**GPIO38** General digital I/O; board loads unverified
09	**3V3** 3.3 V rail; current unknown	**3V3** 3.3 V rail; current unknown	**GPIO35** **memory caution**	**GPIO37** **memory caution**
10	**GND** Ground	**GND** Ground	**GPIO33** **memory caution**	**GPIO34** **memory caution**
11	**VIN** Supply rail; input limits unverified	**VIN** Supply rail; input limits unverified	**GPIO48** General digital I/O; board loads unverified	**GPIO47** General digital I/O; board loads unverified

GPIO19 USB D- and GPIO20 USB D+ are not on these four header columns. TX/RX are interpreted as default GPIO43/44 and are corroborated by the back photo's IO43 TX / IO44 RX labels. [S1, S5]

Unlike the larger white board: GPIO37 and GPIO47 occupy distinct MiniNano positions (D09 and D11). There is no 37/47 label conflict here, but GPIO37 remains a memory-reserved pin on octal-PSRAM hardware. GPIO48 being exposed does not prove it drives the visible RGB-like LED.

Verified chip-family information and cautions

Topic	Reference information
CPU / memory	Dual-core Xtensa LX7, up to 240 MHz; 512 KB SRAM. N16R8 convention denotes 16 MB flash and 8 MB PSRAM; ESP32-S3R8 has embedded octal-SPI PSRAM. Verify actual fitted memory. [S1, S2, S4]
Radio	2.4 GHz 802.11 b/g/n Wi-Fi; Bluetooth 5 LE. No Bluetooth Classic or 5 GHz Wi-Fi. [S4]
Peripherals	GPIO matrix, SPI, I2C, UART, I2S, PWM/RMT, ADC, touch, SD/MMC and USB. Routing is flexible, not unlimited. GPIO22-25 do not exist. [S1, S3, S4]
Analog	ADC1 on GPIO1-10; ADC2 on GPIO11-20; touch on GPIO1-14. Prefer ADC1 for Hall sensors. ADC2 shares hardware with Wi-Fi; use supported driver arbitration and calibration. [S3, S6]
USB	Full-speed USB, 12 Mbit/s; HID/CDC/composite devices via TinyUSB. USB OTG and Serial/JTAG share the internal PHY. [S5]

3.3 V logic - GPIO is not 5 V tolerant. Chip supply recommendations are 3.0-3.6 V; this is not a specification for VIN. Do not apply 5 V to GPIO or 3V3. [S1]

VIN / power - The supplied chart labels two pins VIN, not a verified 5V/12V input range. Regulator part, current capacity, rail ties and USB reverse-current protection are unknown. Do not apply an arbitrary VIN voltage or combine independent power sources without checking the circuit.

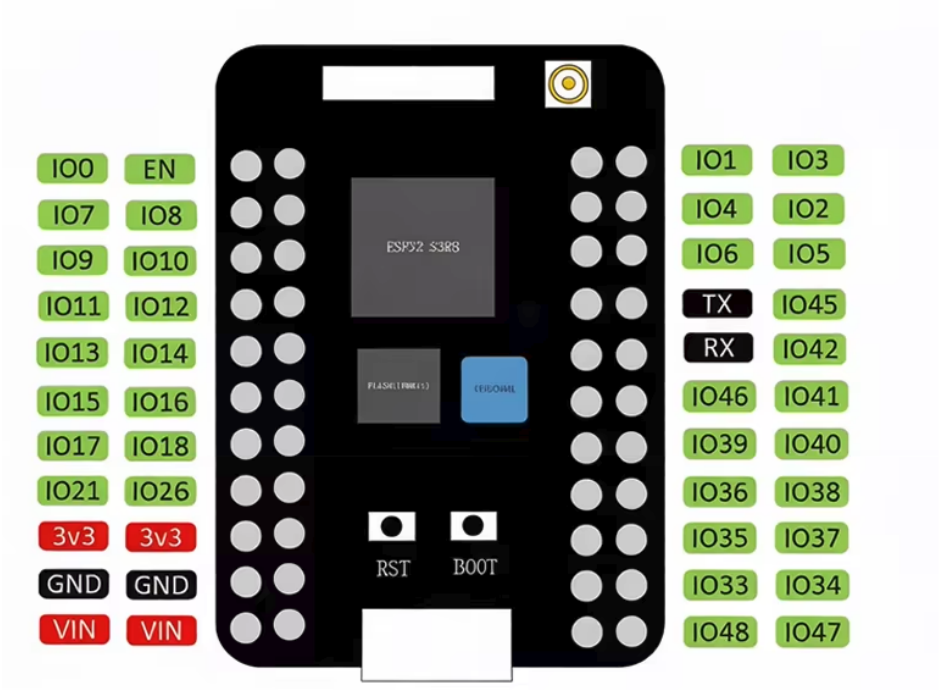
Flash and PSRAM - GPIO26 is a SPI0/1 memory signal. GPIO33-37 carry extra octal data/DQS signals when octal memory is used. The header exposes them, but that does not make them usable as spare I/O on R8 hardware. Reconfiguring them can corrupt memory or crash the device. [S1, S3]

Strapping - GPIO0, GPIO3, GPIO45 and GPIO46 are sampled at reset. GPIO45 may set VDD_SPI voltage; GPIO0 low and GPIO46 low permit joint download boot. Do not add pulls or peripherals that defeat required boot levels. Current chip documentation lists GPIO46 as I/O/T. [S1]

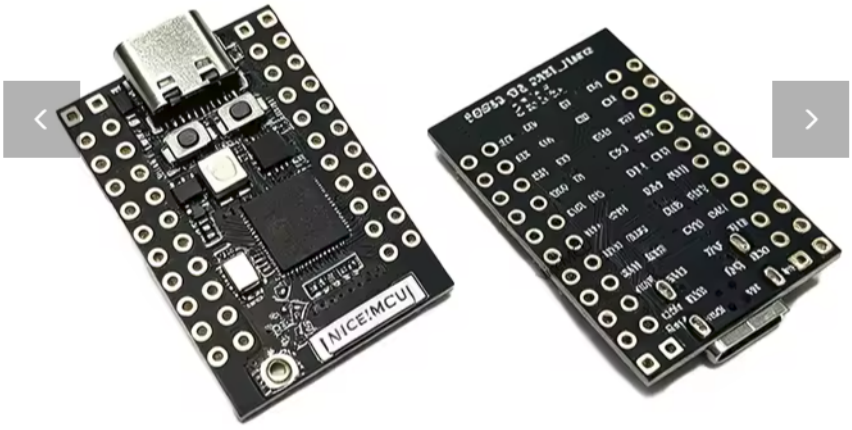
JTAG / UART - GPIO39-42 overlap external JTAG; GPIO43/44 are default UART0 TX/RX. Use the GPIO matrix intentionally if remapping. Avoid conflicts with boot/debug signals. [S1, S3]

LED / RF routing - The photos show an RGB-like LED, ceramic antenna and external RF receptacle. Exact LED GPIO/protocol and antenna selection/matching network are unknown. Do not assume the larger board's GPIO48 LED circuit or simultaneous active antennas.

Source pinout and photographed identity



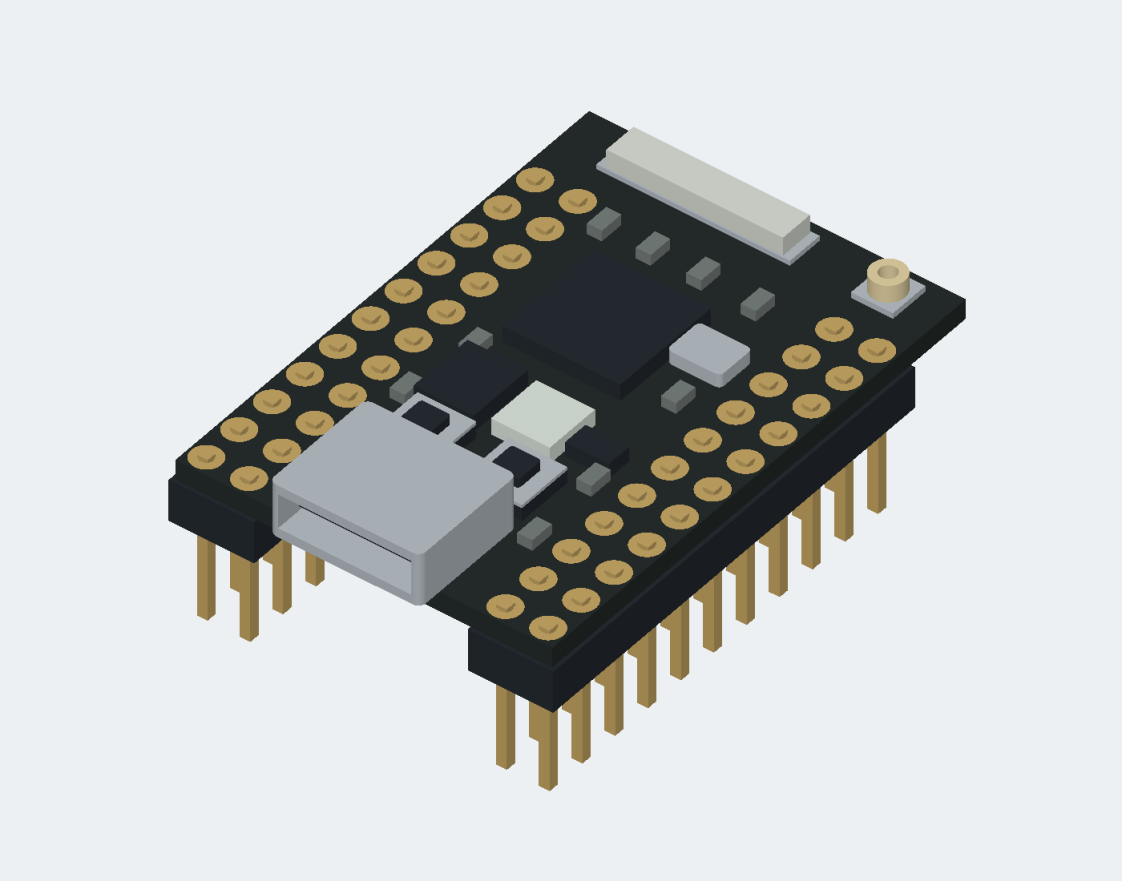
Supplied interface drawing. Its exposed memory GPIOs are not a guarantee of availability with octal PSRAM.



Supplied NiceMCU-branded front/back photograph. The model omits printed branding, tracks, solder and precise small-component leads. It represents component clearance envelopes, not an RF design.

The back uses aliases such as D-numbers in addition to IO labels. This reference consistently uses actual GPIO identifiers from the interface drawing, never an unverified Arduino Nano pin mapping.

Fitted headers, startup and mechanical integration



Four assumed downward 1x11 male strips; dimensions are estimates, not measurements of a supplied header.

Plastic overhang - With 2.54-wide outer strips centered at X +/-10.16, plastic reaches X +/-11.43 versus PCB edges +/-11.20: 0.23 mm per side. Its end also extends 0.23 beyond the USB-end PCB. The fitted envelope is 22.86 x 33.46 x 12.94, not the bare PCB dimensions.

Native USB / recovery - GPIO19 D- and GPIO20 D+ are the chip's native USB signals. The pictured single USB-C suggests native USB but lacks schematic confirmation. Hold BOOT, press/release RST, then release BOOT to request download mode; confirm GPIO46's required level and use a data cable. [S1, S5]

Memory settings - Read flash capacity with esptool, check chip marking and confirm PSRAM initialization in firmware before selecting 16 MB flash partitions or octal-PSRAM settings. N16R8 is a product claim, not an observed memory test.

Orbit Controller use - Use non-strapping ADC1 inputs for Hall joystick signals and verified digital pins for display/buttons. Reserve memory GPIOs and USB, allow for LED/JTAG/UART loads and verify voltage levels. USB HID capability does not itself implement a SpaceMouse protocol. [S3, S5]

Fit and RF - Add clearances for USB cable/plug, header sockets and the external RF cable/connector; keep access to BOOT/RST and space around the ceramic antenna. Antenna keep-out, cable bend radius, thermal performance, power integrity and physical fit must be checked on a real sample.

Sources and limits of this reference

Research date: 1 November 2026. The same verified ESP32-S3 family sources used for the larger board apply to chip capabilities, not to MiniNano-specific routing. No exact MiniNano schematic was found; similarly named Nano/SuperMini board information is not used as a substitute.

[S1] Espressif ESP32-S3 Series Datasheet, v2.2
Chip variants, GPIO functions, boot straps, supply recommendations and package reference.
https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf

[S2] Espressif ESP32-S3-WROOM-1 / WROOM-1U Datasheet, v1.8
N16R8 memory nomenclature only; this model is not claimed to contain that shielded module.
https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

[S3] ESP-IDF ESP32-S3 GPIO and RTC GPIO reference
ADC pin map, GPIO matrix, memory-reserved pins, USB19/20 and strapping summary.
https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/gpio.html

[S4] Espressif ESP32-S3 product overview
CPU, SRAM, Wi-Fi, Bluetooth LE and supported interfaces.
https://www.espressif.com/en/products/socs/esp32-s3

[S5] Espressif ESP-USB USB Device Stack for ESP32-S3
USB D-/D+ routing, full-speed operation, TinyUSB classes and shared PHY limitation.
https://docs.espressif.com/projects/esp-usb/en/latest/esp32s3/usb_device.html

[S6] ESP-IDF ESP32-S3 ADC Oneshot Mode Driver
ADC2/Wi-Fi sharing and driver protection; calibration and error handling.
https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/adc/adc_oneshot.html

Supplied local references

ESP32-S3-N16R8-MiniNano- dimensions.png; pins.png; frontback.png; front.png; back.png (each filename uses that same prefix). Copies are included as reference-dimensions/pins/frontback/front/back.png.

Observed, inferred and unknown

Observed: dimension labels 22.40 x 31.96; 44 pad locations; component appearance; supplied GPIO map. **Inferred:** nominal header grid and offsets; board/connector/component heights; header lengths; USB overhang; pad/drill diameters. **Unknown:** exact schematic, voltage/current limits, actual memory population, LED GPIO, antenna selection, tolerances and physical fitting.

Model scope and checks before manufacture

The CAD exports preserve the modeled solids and nominal envelopes. They omit traces, solder, fine QFN leads, precise USB/RF contacts and antenna matching details; square end pads are simplified. Measure the real board's inferred row gaps, PCB thickness and connector heights; verify power routing and reserved pins; exercise USB and memory; then confirm enclosure and mating-board clearances. This document and the models are mechanical/electrical references, not certified production data.